

What is "The Francisco Problem"?

In traditional smoothing algorithms such as **Absolute Discounting** or **Jelinek-Mercer**, lower-order distributions (such as unigrams in a bigram model) are calculated using **raw maximum likelihood frequencies**.

The fundamental flaw in this approach is that raw frequency does not reflect context versatility. Consider a word like **"Francisco"**:

- In any reasonably sized English corpus, "Francisco" appears frequently, so its raw count $c(\text{"Francisco"})$ is high.
- Because its raw count is high, its raw unigram probability $P_{\text{ML}}(\text{"Francisco"})$ is also high.
- However, "Francisco" almost exclusively occurs following one specific preceding word: **"San"**.

When a model encounters a novel or unseen context (such as predicting the next word in "read ____" or "great ____"), smoothing algorithms steal probability mass from observed n -grams and distribute it using lower-order distributions. Because Absolute Discounting uses raw unigram frequencies for its lower-order term, it assigns a relatively high probability to "Francisco" in completely novel contexts, leading to absurd predictions like "read Francisco" or "great Francisco".

How Kneser-Ney Solves It: Continuation Probability

Kneser-Ney changes the question asked by the lower-order model:

- **Standard Unigram:** *"How frequent is this word overall in the corpus?"*
- **Kneser-Ney Lower-Order:** *"In how many **different preceding contexts** does this word appear as a novel continuation?"*

Instead of raw counts, Kneser-Ney uses a **continuation count** $N_{1+}(\cdot w)$, which tallies the number of unique preceding words w_{i-1} that w follows:

$$N_{1+}(\cdot w) = |\{w_{i-1} : c(w_{i-1}w) > 0\}|$$

The normalized **continuation probability** $P_{\text{cont}}(w)$ is then:

$$P_{\text{cont}}(w) = \frac{N_{1+}(\cdot w)}{\sum_{w'} N_{1+}(\cdot w')}$$

Because "Francisco" only ever follows "San", its continuation count is 1, giving it a low continuation probability. Conversely, a versatile word like "book" follows dozens of different words ("a book" , "the book" , "my book"), giving it a high continuation probability.

Toy Example 1: Bigram Level

Suppose our training corpus contains the following 4 sentences:

1. <s> I visit San Francisco </s>
2. <s> They visit San Francisco </s>
3. <s> She read a book </s>
4. <s> He read the book </s>

Count Statistics:

- **"Francisco"**: Raw count $c(\text{"Francisco"}) = 2$. It appears after only 1 unique word ("San") $\rightarrow N_{1+}(\text{"Francisco"}) = 1$.
- **"book"**: Raw count $c(\text{"book"}) = 2$. It appears after 2 unique words ("a" and "the") $\rightarrow N_{1+}(\text{"book"}) = 2$.

Evaluating an Unseen Bigram Context: "want ____"

Since "want" was never seen in training, the model must rely on the lower-order distribution to predict the next word:

- **Absolute Discounting**: Uses raw unigrams $P_{\text{ML}}(w)$. Because $c(\text{"Francisco"}) = c(\text{"book"}) = 2$, both words receive equal fallback probability. If "Francisco" had appeared 10 times after "San", Absolute Discounting would wrongly predict "want Francisco" over "want book" .
 - **Kneser-Ney**: Uses continuation probability $P_{\text{cont}}(w)$. Since $N_{1+}(\text{"book"}) = 2$ and $N_{1+}(\text{"Francisco"}) = 1$, Kneser-Ney assigns twice as much lower-order probability to "book" as it does to "Francisco" .
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Toy Example 2: Extending to the Trigram Level

In a **trigram model**, we predict word w_i given a 2-word history (w_{i-2}, w_{i-1}). Kneser-Ney applies continuation counts **recursively** to every intermediate lower-order distribution.

When a trigram (w_{i-2}, w_{i-1}, w_i) is unseen, the model interpolates down to a **smoothed bigram distribution** $P_{\text{KN}}(w_i | w_{i-1})$. To prevent the "Francisco problem" at the bigram level, Kneser-Ney replaces raw bigram counts $c(w_{i-1}w_i)$ with **trigram continuation counts** $N_{1+}(\cdot w_{i-1}w_i)$:

$$N_{1+}(\cdot w_{i-1}w_i) = |\{w_{i-2} : c(w_{i-2}w_{i-1}w_i) > 0\}|$$

This counts **how many distinct 2-word contexts** w_{i-2} **can precede the bigram** $w_{i-1}w_i$.

Trigram Corpus Scenario:

Suppose we want to predict the next word for an unseen trigram context: "likes the ____".

The model drops the oldest word ("likes") and evaluates the lower-order bigram distribution for histories starting with "the".

Compare two candidate bigrams following "the" :

1. Bigram A: "San Francisco"

- Suppose in training, "San Francisco" always appeared in the exact same 3-gram context: "visit San Francisco".
- Preceding words w_{i-2} before "San Francisco": Only {"visit"}.
- **Trigram Continuation Count:** $N_{1+}(\cdot \text{"San Francisco"}) = 1$.

2. Bigram B: "green book"

- Suppose in training, "green book" appeared across several distinct 3-gram contexts: "read green book", "bought green book", and "saw green book".
- Preceding words w_{i-2} before "green book": {"read", "bought", "saw"}.
- **Trigram Continuation Count:** $N_{1+}(\cdot \text{"green book"}) = 3$.

Result at the Trigram Level:

When backing off from the unseen trigram "likes the ____", Kneser-Ney assigns a significantly higher bigram probability to "green book" than to "San Francisco" because "green book" demonstrated higher versatility across 3-gram contexts.

But aren't we calculating only next word? Why are u predicting next 2 words in your example?

You are 100% correct! Language models always predict **exactly one single next word** (w_i) at a time [12–16, 311–312].

The mention of two-word phrases like "San Francisco" or "green book" in the previous discussion was shorthand for the resulting bigram formed with the context, which created understandable confusion.

Clearing Up the Example Step-by-Step

In a **trigram model**, we are predicting the **single next word** w_i given a 2-word history (w_{i-2}, w_{i-1}) [15–16, 312–313]:

- **History Context:** $w_{i-2} = \text{"likes"}$, $w_{i-1} = \text{"the"}$
- **Target to Predict:** One single word w_i

Comparing the Two SINGLE Word Candidates

We are choosing between candidate target words for w_i :

Candidate Target Word (w_i)	Resulting 3-word Sequence	Formed Bigram ($w_{i-1}w_i$)	Kneser-Ney Continuation Evaluation
Candidate 1: $w_i = \text{"book"}$	"likes the book"	"the book"	High versatility: "book" follows many different preceding words ("a book" , "the book" , "my book" , "green book") [75–77, 318–319].
Candidate 2: $w_i = \text{"Francisco"}$	"likes the Francisco"	"the Francisco"	Low versatility: "Francisco" in English text almost exclusively follows the single word "San" [75, 317–318].

How Kneser-Ney Evaluates the Single Target Word w_i

When the 3-gram "likes the ____" is unseen in training, Kneser-Ney drops the oldest word ("likes") and evaluates the lower-order continuation probability for the single target word w_i [79, 85–86]:

1. **For** $w_i = \text{"Francisco"}$:

Kneser-Ney checks how many unique preceding words w_{i-1} appear before "Francisco" across the corpus. Since that count $N_{1+}(\cdot \text{"Francisco"})$ is essentially 1 (only "San"), its continuation probability $P_{\text{cont}}(\text{"Francisco"})$ is extremely small [75–77, 318–319].

2. **For** $w_i = \text{"book"}$:

Kneser-Ney checks how many unique preceding words w_{i-1} appear before "book" . Since "book" appears after many different words, its continuation count $N_{1+}(\cdot \text{"book"})$ is high, giving $P_{\text{cont}}(\text{"book"})$ a much higher score [75–77, 318–319].

The Final Prediction

The model assigns a high probability to the single target word "book" , producing the complete sentence:

"likes the book"